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High heat transfer cryogenic bearing

Abstract

A superconducting motor including a rotor, a heat sink, and a bearing. The rotor includes a superconducting winding formed from a superconducting material. The heat sink is positioned within the superconducting motor to absorb heat produced by the superconducting motor. The bearing includes an inner race, an outer race, and a rolling element. The bearing rotatably supports the rotor and is thermally connected to each of the rotor and the heat sink to transfer heat from the rotor to the heat sink. The rolling element is formed from a material having, at 25 K, a thermal conductivity of 20 W/mK or more and an ultimate tensile strength of 650 MPa or more. The rolling element may be formed from a copper alloy.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a bearing for a rotating device, particularly a bearing used in an electric motor.

BACKGROUND

(2) Propulsion systems for commercial aircraft typically include one or more aircraft engines, such as turbofan jet engines. These engines may be powered by aviation turbine fuel, which is typically a combustible hydrocarbon liquid fuel, such as a kerosene-type fuel, having a desired carbon number and carbon-to-hydrogen ratio. Such fuel produces carbon dioxide upon combustion. Using electrical propulsion or hybrid electrical propulsion may reduce or eliminate such carbon dioxide emissions, and such electrical propulsion systems would use an electric motor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features and advantages of the present disclosure will be apparent from the following description of various exemplary aspects, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 is a schematic plan view of an aircraft having an electrical propulsion system.

(3) FIG. 2 is a schematic, cross-sectional view of an electric motor of the electrical propulsion system shown in FIG. 1.

(4) FIG. 3 is a cross-sectional view of a portion of the electric motor having a bearing according to

an aspect of the present disclosure.

(5) FIG. 4 is a cross-sectional view of a portion of the electric motor having a bearing according to an aspect of the present disclosure.

(6) FIG. 5 is a cross-sectional view of a bearing according to an aspect of the present disclosure that may be used in the electric motor.

(7) FIG. 6 is a cross-sectional view of a portion of the electric motor having a bearing according to an aspect of the present disclosure.

(8) FIG. 7 is a cross-sectional view of a bearing according to an aspect of the present disclosure that may be used in the electric motor.

DETAILED DESCRIPTION

(9) Features, advantages, and aspects of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

(10) Various aspects are discussed in detail below. While specific aspects are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and the scope of the present disclosure.

(11) The terms “forward” and “aft” refer to relative positions within the propulsion system or vehicle (e.g., aircraft), and refer to the normal operational attitude of the propulsion system or vehicle.

(12) The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting, as well as, indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.

(13) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(14) Here and throughout the specification and claims, range limitations are combined and interchanged. Such ranges are identified and include all the sub-ranges contained therein, unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(15) As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to the longitudinal centerline of the motor or the rotational axis of the bearing, which in aspects herein are coincident with each other. The terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the longitudinal centerline of the motor or the rotational axis of the bearing. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the longitudinal centerline of the motor or the rotational axis of the bearing.

(16) References to “inner” and “outer” when discussed in the context of radial directions refer to positions relative to the longitudinal centerline of the motor or the rotational axis of the bearing. References to “inner” and “outer” when discussed in the context of axial directions are taken in context of a meridian plane of the component being discussed, such as the bearing. Such meridian planes extend substantially perpendicular to the longitudinal centerline of the motor or the rotational axis of the bearing.

(17) As noted above, aircraft engines may be powered by aviation turbine fuel. Aviation turbine fuel is typically a combustible hydrocarbon liquid fuel that produces carbon dioxide upon combustion. Efforts are being made to reduce the amount of carbon dioxide produced by aircraft. An electric motor may be used in the propulsion system for the aircraft as either part of a hybrid electrical engine propulsion system or even an electrified aircraft. The electric motor may be used to reduce or even to eliminate carbon dioxide emissions from commercial aircraft.

(18) Electric motors dissipate some energy as heat and are thus less than one hundred percent efficient. These energy losses may be caused by, for example, mechanical friction, electrical resistance with the windings, eddy current effects, and hysteresis. One way to improve the efficiency of the electric motor is to use relatively large windings, as such windings have relatively low electrical resistance. However, such large windings are undesirable in aircraft because the larger the winding, the more the electric motor weighs. In aircraft, propulsion systems with a high power to weight ratio are desired, and thus such large electric motors are undesirable.

(19) Another way to increase the efficiency of the electric motor is to use superconducting materials (superconductors) for the windings. An ordinary metallic conductor has some gradual decrease in electrical resistance with a decrease in temperature, but even at temperatures near absolute zero, an ordinary metallic conductor may maintain significant electrical resistance. Superconducting materials, however, have a characteristic critical temperature below which the resistance drops abruptly to zero. An electric motor with windings formed of superconducting materials can thus operate at high efficiencies with relatively small windings, when the electric motor is operated at temperatures below the critical temperature of the superconductor. Superconducting motors that have windings made from superconducting materials and are operated below the critical temperature of the superconductor may thus have power to weight ratios that are desirable for use in aircraft propulsion.

(20) Thermal management of the motor is necessary in order to maintain the motor, and, thus, the windings, below the critical temperature. Critical temperatures for superconductors are in the cryogenic range and, thus, the operating temperature of superconducting motors is preferably kept at temperatures less than fifty kelvin, such as thirty kelvin or less, or such as twenty-five kelvin or less. Even these superconducting electric motors will generate heat as noted above. The superconducting electric motors discussed herein utilize a high heat transfer bearing to transfer heat from the rotor to the stator of the motor where the heat can be removed from the motor by a suitable cooling system. The high heat transfer bearings and bearing assemblies discussed herein thus help to maintain the temperature of the motor at appropriate operating temperatures.

(21) The bearings discussed herein are particularly suitable for use in superconducting electric motors for use in an aircraft. FIG. 1 is a schematic plan view of an aircraft 10 that implements an electric motor with various preferred high heat transfer bearings and bearing assemblies discussed herein. The aircraft 10 includes a fuselage 12, wings 14 attached to the fuselage 12, and an empennage 16. The aircraft 10 also includes a propulsion system 20 that produces a propulsive thrust required to propel the aircraft 10 in flight, during taxiing operations, and the like. The propulsion system 20 includes at least one electric propulsor 30. The electric propulsor 30 includes a motor 100 coupled to a rotating airfoil assembly 32. The motor 100 shown in FIG. 1 is an electric motor and a torque producing system used to produce torque. The motor 100 includes an output shaft 124 that is mechanically coupled to the rotating airfoil assembly 32. The output shaft 124 transmits the torque produced by the motor 100 from the motor 100 to the rotating airfoil assembly 32.

(22) The rotating airfoil assembly 32 includes a plurality of airfoils rotating about a central axis to produce the thrust used to propel the aircraft 10. In the aspect shown in FIG. 1, the rotating airfoil assembly 32 is a fan and the rotating airfoils are fan blades. The fan (rotating airfoil assembly 32) shown in FIG. 1 is a ducted fan with an annular fan casing 34 (or an outer nacelle) provided circumferentially surrounding the fan (rotating airfoil assembly 32) and/or at least a portion of the motor 100. The motor 100 discussed herein, however, may be coupled to other rotating airfoil assemblies such as unducted fans and propellers. Such propellers or fans may be arranged in a pusher configuration, where the rotating airfoil assembly is forward of the motor 100, or a puller configuration, where the rotating airfoil assembly is aft of the motor 100. The motor 100 may be coupled to a plurality of rotating airfoil assemblies, such as a plurality of fans arranged in a contra-rotating arrangement, for example.

(23) The propulsion system **20** for the aircraft **10** shown in FIG. **1** includes a pair of electric propulsors **30** attached to the fuselage **12**, such as proximate the empennage **16**, one on the port side of the fuselage **12** and one on the starboard side of the fuselage **12**. In other aspects, the electric propulsor **30** may have alternative configurations and be coupled to other portions of the aircraft **10**. For example, the electric propulsor **30** may additionally or alternatively include one or more aspects coupled to other parts of the aircraft **10**, such as, for example, one of the wings **14** by a pylon in an under-wing configuration.

(24) The aircraft **10** shown in FIG. **1** is an electrified aircraft, and the propulsion system **20** includes a source of electricity (an electrical power source). In this aspect, the propulsion system **20** includes a plurality of sources of electricity, a fuel cell stack **22**, and a battery **24**. Each of the fuel cell stack **22** and the battery **24** is electrically coupled to an electrical bus **26** used to distribute electricity throughout the aircraft **10**, including to the motors **100** of the electric propulsors **30**. Each of the fuel cell stack **22** and the battery **24** is thus electrically coupled to the motor **100** via the electrical bus **26**.

(25) The fuel cell stack **22** may be a hydrogen fuel cell stack, and a fuel tank **28** storing hydrogen, such a diatomic hydrogen, may be fluidly coupled to the fuel cell stack **22** to produce electricity. The fuel tank **28** may be configured to hold the hydrogen fuel at least partially within the liquid phase. To store the hydrogen fuel in the liquid phase, the hydrogen fuel is stored in the fuel tank **28** at very low (cryogenic) temperatures. For example, the hydrogen fuel may be stored in the fuel tank **28** at about -253 degrees Celsius (twenty kelvin) or less at atmospheric pressure, or at other temperatures and pressures to maintain the hydrogen fuel in the liquid phase. In some aspects, the hydrogen fuel may be stored in the fuel tank **28** at temperatures from -259 degrees Celsius (fourteen kelvin) to -243 degrees Celsius (thirty kelvin), and more preferably, from -253 degrees Celsius (twenty kelvin) to -243 degrees Celsius (thirty kelvin). As shown in FIG. **1**, at least a portion of the fuel tank **28** is located in the fuselage **12**, and, in this aspect, entirely within the fuselage **12**.

(26) The electric propulsor **30** may also include a motor control unit **40** (motor controller). The motor control unit **40** is connected to the motor **100** and is configured to operate various aspects of the motor **100**, including, in some aspects, regulating an electrical input and a rotational output of the motor **100**. In this aspect, the motor control unit **40** is a computing device having one or more processors **42** and one or more memories **44**. The processor **42** can be any suitable processing device, including, but not limited to, a microprocessor, a microcontroller, an integrated circuit, a logic device, a programmable logic controller (PLC), an application-specific integrated circuit (ASIC), and/or a Field Programmable Gate Array (FPGA). The memory **44** can include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, a computer-readable non-volatile medium (e.g., a flash memory), a RAM, a ROM, hard drives, flash drives, and/or other memory devices.

(27) The memory **44** can store information accessible by the processor **42**, including computer-readable instructions that can be executed by the processor **42**. The instructions can be any set of instructions or a sequence of instructions that, when executed by the processor **42**, causes the processor **42** and the motor control unit **40** to perform operations. In some aspects, the instructions can be executed by the processor **42** to cause the processor **42** to complete any of the operations and functions for which the motor control unit **40** is configured, as will be described further below. The instructions can be software written in any suitable programming language, or can be implemented in hardware. Additionally, and/or alternatively, the instructions can be executed in logically and/or virtually separate threads on the processor **42**. The memory **44** can further store data that can be accessed by the processor **42**.

(28) The technology discussed herein makes reference to computer-based systems and actions taken by, and information sent to and from, computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety

of possible configurations, combinations, and divisions of tasks and functionality between components and among components. For instance, processes discussed herein can be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel. (29) Although the aircraft **10** shown in FIG. **1** is an airplane, the disclosure herein may also be applicable to other aircraft, including, for example, helicopters and unmanned aerial vehicles (UAV). Further, although not depicted herein, in other aspects, the motor **100** may be used in other applications, including, for example, other commercial and industrial motor applications. The motor **100** may thus be connected to drive other mechanical elements (e.g., axles).

(30) FIG. **2** is a schematic, cross-sectional view of one of the motors **100** shown in FIG. **1**. The motor **100** includes a stator **110** and a rotor **120** rotatable about a longitudinal axis **101** of the motor **100**. The motor **100** includes a radial direction R and an axial direction A. In this aspect, the stator **110** is positioned radially inward of the rotor **120**, but in other aspects, the rotor **120** may be positioned radially inward of the stator **110**. Each of the stator **110** and the rotor **120** includes windings electrically connected to the electricity source, such as the fuel cell stack **22** or the battery **24**, to receive electrical current from the electricity source. One of the windings is an armature winding **112**. The armature winding **112** is formed as part of the stator **110** in this aspect. The other winding is a field winding **122**, and, in this aspect, the field winding **122** is formed as part of the rotor **120**. The output shaft **124** is connected to the rotor **120** and, as electrical current is passed through the windings (the armature winding **112** and the field winding **122**), the rotor **120** rotates, producing the torque transmitted by the output shaft **124**. As discussed above, the motor **100** is a superconducting motor and the armature winding **112** and the field winding **122** are each formed from a superconducting material. Suitable superconducting materials include, for example, niobium alloys and other niobium compounds, vanadium alloys and other vanadium compounds, magnesium diboride, and cuprates. Examples of such superconducting materials (and others) are listed in Table 1, together with the critical temperature for each material.

(31) TABLE-US-00001
 TABLE 1 Critical Superconductor Temperature (K)
 Niobium Alloys and Compounds
 NbTi 9.3 Nb.sub.3Sn 18 Nb.sub.3Al 18.9 Nb.sub.3Ga 20.3
 Nb.sub.3(Al.sub.75Ge.sub.25) 20.5 Nb.sub.3Ge 23 NbN 16
 Vanadium Alloys and Compounds
 V.sub.3Ga 15 V.sub.3Si 16 V.sub.2(Hf, Zr) 10.1
 Cuprates
 La.sub.1.85Sr.sub.0.25CuO.sub.4-δ 40
 YBa.sub.2Cu.sub.3O.sub.7-δ 90 Bi.sub.2Sr.sub.2CaCu.sub.2O.sub.8-δ 90 (Bi, Pb).sub.2Sr.sub.2CaCu.sub.2O.sub.8+ 110
 Tl.sub.2Ba.sub.2CaCu.sub.2O.sub.8+δ 110
 Tl.sub.2Ba.sub.2CaCu.sub.3O.sub.10-δ 125 HgBa.sub.2CaCu.sub.3O.sub.8+δ 133
 Others
 PbMo.sub.6S.sub.8 15 MgB.sub.2 39

(32) As can be seen in Table 1, these critical temperatures are in the cryogenic range and, thus, the operating temperature of the motor **100** is preferably kept below these temperatures and, more preferably, well below these temperatures to avoid the risk of the windings rising to a temperature where the windings lose their super conductivity. The operating temperature of the motor **100** may be kept at temperatures less than sixty kelvin (60 K), such as fifty kelvin (50 K) or less, such as thirty kelvin (30 K) or less, or such as twenty-five kelvin (25 K) or less. The motor **100** includes a heat sink positioned within the motor **100** to remove heat from the motor **100**. In this aspect, the heat sink is a cooling coil **130** thermally coupled to the stator **110**. A cooling fluid (coolant) suitable to maintain the motor **100** at the operating temperatures discussed above is passed through the cooling coil **130**. Suitable coolants include, for example, liquid hydrogen or liquid helium. For example, the liquid hydrogen used as the fuel for the fuel cell stack **22** may be used as the coolant. The cooling coil **130** includes an inlet **132** fluidly coupled to a coolant source such as the fuel tank **28**. The coolant passes through the cooling coil **130** receiving heat from the stator **110** and cooling the stator **110** and then exits the cooling coil **130** via an outlet **134**.

(33) To further help maintain the operating temperature of the motor **100**, the motor may be

operated in a vacuum. The motor **100** thus further includes a vacuum vessel **140** including a cavity **142**. The vacuum is drawn in the cavity **142** and the windings (the armature winding **112** and the field winding **122**) may be located within the cavity **142**.

(34) The rotor **120** is supported relative to the stator **110** by a bearing **152**. The bearing **152** is preferably suitable for use in the operating temperatures discussed herein and is thus a cryogenic bearing. Heat may also be generated in the rotor **120**. In this aspect, the heat sink (cooling coil **130**) is located within the stator **110** and, thus, the heat from the rotor **120** should be transferred from the rotor **120**. The rotor **120** is, thus, thermally coupled to the stator **110** by the bearing **152**, and the bearing **152** is, thus, preferably a high-heat transfer bearing that efficiently transfers heat from the rotor **120** to the stator **110**. In aspects, such as those discussed herein, the bearing **152** may be formed as part of a high-heat transfer bearing assembly **150**. In this aspect, the motor **100** includes a plurality of high-heat transfer bearing assemblies **150**, each having the bearing **152**. In FIG. 2, for example, a high-heat transfer bearing assembly **150** is disposed on each end of the stator **110** and rotor **120**. More specifically, a high-heat transfer bearing assembly **150** is disposed on each end of the armature winding **112** and the field winding **122**. The aspects of the high-heat transfer bearing assembly **150** and the bearing **152** discussed below are described with reference to one of the high-heat transfer bearing assemblies **150** and bearings **152**, but such descriptions apply to the other high-heat transfer bearing assemblies **150** and bearings **152** in the motor **100**.

(35) FIG. 3 is a cross-sectional view of a portion of the motor **100** having a bearing **200** that may be used as a bearing **152** of the motor **100** shown in FIG. 3. The bearing **200** of this aspect is a roller bearing that includes a plurality of rolling elements **202** positioned between an inner race **204** (an inner ring in the aspect shown in FIG. 3) and an outer race **206** (an outer ring in the aspect shown in FIG. 3).

(36) In FIG. 3, the rolling element **202** is a ball, but other suitable rolling elements **202** may be used such as a roller and, more specifically, a cylindrical roller, for example. In some aspects, the rolling elements **202** may be positioned relative to each other by a retainer **208** (also referred to as a cage). The rolling elements **202** are arrayed in a circumferential direction of the bearing **200**, and the retainer **208** thus spaces the rolling elements **202** apart from each other in the circumferential direction. The circumferential direction extends in a direction rotating about the rotational axis **201** of the bearing **200**, which, in this aspect, is coincident with the longitudinal axis **101** of the motor **100**. Each of the inner race **204** and the outer race **206** may have a circumferential groove to guide the rolling elements **202** as the rolling elements **202** rotate in the bearing **200**.

(37) In this aspect, the inner race **204** is connected to the stator **110** and the outer race **206** is connected to the rotor **120**. As noted above, the bearing **200** is a bearing configured to transfer heat from the rotor **120** to the stator **110**, and, thus, the inner race **204** is thermally coupled to the stator **110** and the outer race **206** is thermally coupled to the rotor **120**. In this aspect, the inner race **204** is directly thermally coupled to the stator **110**, such as by contacting the stator **110**, and the outer race **206** is directly thermally coupled to the rotor **120**, such as by contacting the **120**. The outer race **206** includes two axial surfaces **206a** and an outer circumferential surface **206b**. The outer circumferential surface **206b** of the outer race **206** is directly attached to the rotor **120**. Similarly, the inner race **204** includes two axial surfaces **204a** and an inner circumferential surface **204b**. The inner circumferential surface **204b** of the inner race **204** is directly attached to the stator **110**.

(38) The field winding **122** and the rotor **120** should have from one to ten watts of cooling to maintain the field winding **122** and the rotor **120** at operating temperatures when the field winding **122** is below its critical temperature, and, thus, superconducting. Stainless steel is a material that may be used for the components of a cryogenic bearing, but the thermal conductivities of bearings made from such materials, however, is not high enough to provide the cooling capacity for the rotor **120**. For example, 316 stainless steel has a thermal conductivity of two watts per meter kelvin (2 W/mK) and a thermal resistance of seven hundred fifty kelvin per watt (750 K/W) at twenty-five kelvin (25 K). The bearing **200** of this aspect is formed of materials that have a higher thermal

conductivity at cryogenic temperatures.

(39) Preferably, the bearing **200** and, more specifically, the rolling elements **202**, the inner race **204**, and the outer race **206** are each formed of a material that has a thermal conductivity of that is one or two orders of magnitude higher than stainless steel at twenty-five kelvin (25 K), such as, for example twenty watts per meter kelvin (20 W/mK) or more at twenty-five kelvin (25 K), more preferably a thermal conductivity of one hundred watts per meter kelvin (100 W/mK) or more at twenty-five kelvin (25 K), and even more preferably a thermal conductivity of two hundred watts per meter kelvin (200 W/mK) or more at twenty-five kelvin (25 K). Suitable materials include copper alloys. As used herein, a copper alloy is an alloy where the weight percentage of the copper in the alloy is greater than the weight percentage of any other component of the alloy, based on the total weight of the alloy. As these materials are used in the contact surfaces of the bearing **200**, high-strength copper alloys are preferably used. As used herein, high-strength alloys, such as the high-strength copper alloys, are alloys having an ultimate tensile strength of six hundred ninety megapascals (650 MPa) at twenty-five kelvin (25 K). Suitable high-strength copper alloys include, for example, beryllium copper alloys or copper nickel alloys. The bearing **200** is not limited to copper alloys, and other materials, such as other alloys, meeting both strength and thermal conductivity conditions discussed above, may be used. The bearing **200** and, more specifically, the rolling elements **202**, the inner race **204**, and the outer race **206** thus may each be formed of a material having, at twenty-five kelvin (25 K), both a thermal conductivity of twenty watts per meter kelvin (20 W/mK) or more and an ultimate tensile strength of six hundred ninety megapascals (650 MPa) or more, more preferably both a thermal conductivity of one hundred watts per meter kelvin (100 W/mK) or more and an ultimate tensile strength of six hundred ninety megapascals (650 MPa) or more, and even more preferably both a thermal conductivity of two hundred watts per meter kelvin (200 W/mK) or more and an ultimate tensile strength of six hundred ninety megapascals (650 MPa) or more.

(40) With the bearing **200** and, more specifically, each of the rolling elements **202**, the inner race **204**, and the outer race **206**, formed of the materials discussed above, such as, preferably, the high-strength copper alloys, the bearing **200** thermally couples the rotor **120** to the stator **110** to conduct heat generated in the rotor **120** to the stator **110** and the cooling coil **130** (heat sink) located therein as indicated by the arrows in FIG. 3. In some aspects, each of the rolling elements **202**, the inner race **204**, and the outer race **206** is formed from a copper alloy having the same primary alloying elements and, preferably, may be formed from the same copper alloy.

(41) FIG. 4 is a cross-sectional view of a portion of the motor **100** having a bearing assembly **210** that may be used as the high-heat transfer bearing assembly **150**. The bearing assembly **210** of this aspect includes the bearing **200** discussed above with reference to FIG. 3. The description of the bearing **200** and the motor **100** also applies to this aspect, and the same reference numerals will be used for components the bearing **200** and the motor **100** as in the discussion above. A detailed description of these components is omitted here.

(42) In the aspect shown in FIG. 3, the bearing **200** is directly thermally coupled to the stator **110** and the rotor **120**. In the applications (e.g., aircraft applications) discussed herein, the stator **110** and the rotor **120** may be made from high-strength, light-weight materials. For example, the superconducting windings, more specifically, the armature winding **112** and the field winding **122** may be embedded in carbon fiber to form the stator **110** and the rotor **120**, respectively. Such materials have a relatively low thermal conductivity. To improve the heat transfer from the rotor **120** to the stator **110**, the bearing assembly **210** of this aspect (FIG. 4) includes at least one bearing housing. In this aspect, the bearing **200** is located within a stator bearing housing **212** and a rotor bearing housing **214**. The stator bearing housing **212** and the rotor bearing housing **214** may be formed from a high thermal conductivity material such as the copper alloys discussed above for the bearing **200**. In some aspects, each of the stator bearing housing **212** and the rotor bearing housing **214** are formed from a copper alloy having the same primary alloying elements as the bearing **200**

and, preferably, may be formed from the same copper alloy. The stator bearing housing **212** and the rotor bearing housing **214** may be solid.

(43) The rotor bearing housing **214** is connected to the rotor **120**, and, in this aspect, the rotor bearing housing **214** is directly attached to one end of the rotor **120**. The rotor bearing housing **214** may be a heat sink for the rotor **120** and, more specifically, the field winding **122**. In addition, the rotor bearing housing **214** may have a larger surface area in contact with the bearing **200**, providing a larger heat transfer surface for heat to flow from the rotor **120** to the bearing **200** than in the aspect shown in FIG. 3. In the aspect shown in FIG. 4, the rotor bearing housing **214** has a cross-sectional area that is a U-shape with two axial end portions **214a** extending radially inward on either axial side of the bearing **200**. The bearing **200** is thus located axially between the axial end portions **214a** of the rotor bearing housing **214**. Each axial end portion **214a** includes an inner axial surface **214b** facing the bearing **200** and, more specifically, the outer race **206**. The axial surfaces **206a** (shown in FIG. 3) of the outer race **206** abut the inner axial surfaces **214b** of the rotor bearing housing **214**. In addition, the rotor bearing housing **214** includes an inner circumferential surface **214c** facing the bearing **200** and, more specifically, the outer race **206**. The outer circumferential surface **206b** (shown in FIG. 3) of the outer race **206** abuts the inner circumferential surface **214c** of the rotor bearing housing **214**.

(44) The stator bearing housing **212** of this aspect is positioned to oppose the rotor bearing housing **214** with the bearing **200** therebetween. The stator bearing housing **212** is thus radially inward of the bearing **200** and the rotor bearing housing **214** is radially outward of the bearing **200**. The stator bearing housing **212** is a mirror image of the rotor bearing housing **214** and includes two axial end portions **212a** extending radially inward on either axial side of the bearing **200**. The bearing **200** is thus located axially between the axial end portions **212a** of the stator bearing housing **212**. Each axial end portion **212a** includes an inner axial surface **212b** facing the bearing **200** and, more specifically, the inner race **204**. The axial surfaces **204a** (shown in FIG. 3) of the inner race **204** abut the inner axial surfaces **212b** of the stator bearing housing **212**. In addition, the stator bearing housing **212** includes an outer circumferential surface **212c** facing the bearing **200** and, more specifically, the inner race **204**. The inner circumferential surface **204b** (shown in FIG. 3) of the inner race **204** abuts the outer circumferential surface **212c** of the stator bearing housing **212**. The stator bearing housing **212** is connected to the stator **110**. In this aspect, the stator bearing housing **212** is directly attached to one end of the stator **110**. The stator bearing housing **212** thus includes a large surface area for heat transfer from the bearing **200** to receive heat from the field winding **122**. In some aspects, the stator bearing housing **212** may be in direct contact with the cooling coil **130** to transfer the heat from the bearing **200** to the cooling coil **130**.

(45) FIG. 5 is a cross-sectional view of a bearing assembly **220** that may be used as the high-heat transfer bearing assembly **150**. The bearing assembly **220** of this aspect includes the bearing **200** discussed above with reference to FIG. 3. The description of the bearing **200** also applies to this aspect, and the same reference numerals will be used for components as in the discussion above. A detailed description of these components is omitted here. The bearing assembly **220** is also similar to the bearing assembly **210** discussed above with reference to FIG. 4. The same reference numerals will be used for components of the bearing assembly **220** of this aspect that are the same as or similar to the components of the bearing assembly **210** discussed above. The description of these components above also applies to this aspect, and a detailed description of these components is omitted here.

(46) The bearing assembly **220** of this aspect also includes a stator bearing housing **222** and a rotor bearing housing **224** that have the same shape and positioning as the stator bearing housing **212** and the rotor bearing housing **214** discussed above. In this aspect, however, each of the stator bearing housing **222** and the rotor bearing housing **224** has a cavity **226** formed therein. The cavity **226** may be filled with a heat transfer fluid suitable for use in the operating temperatures of the motor **100** discussed herein. Suitable heat transfer fluids include, for example, hydrogen or helium.

Helium and, more specifically, helium gas may be preferred when the motor **100** operates at the lower end of the operating temperatures discussed herein, such as around twenty kelvin (20 K) or lower such as around ten kelvin (10 K), as temperature fluctuations could cause freezing or so-called slushy hydrogen that would inhibit heat transfer. The heat transfer fluid may thus comprise helium, may consist essentially of helium, or may consist of helium. The use of the heat transfer fluid, such as helium, results in the bearing assembly **220** and, more specifically, the stator bearing housing **222** and the rotor bearing housing **224** of this aspect having a higher heat capacity and thermal conductivity than the solid stator bearing housing **212** and rotor bearing housing **214** discussed above with reference to FIG. 4.

(47) To facilitate heat transfer between the heat transfer fluid and the bearing **200**, each of the stator bearing housing **222** and the rotor bearing housing **224** includes a mesh **228** in a central portion of the cavity **226** between the axial end portions **212a**, **214a**. The mesh **228** is attached to the outer circumferential surface **212c** of the stator bearing housing **222** or the inner circumferential surface **214c** of the rotor bearing housing **224** to provide conductive heat transfer to the inner race **204** or the outer race **206**, respectively. The mesh **228** is thermally coupled to a surface of the rotor bearing housing **224** or the cavity **226** contacting the bearing **200**.

(48) FIG. 6 is a cross-sectional view of a portion of the motor **100** having a bearing assembly **230** that may be used as the high-heat transfer bearing assembly **150**. The bearing assembly **230** of this aspect includes the bearing **200** discussed above with reference to FIG. 3 and the bearing assembly **210**, specifically, the stator bearing housing **212** and the rotor bearing housing **214** of the aspect discussed above with reference to FIG. 4. The description of the motor **100**, the bearing **200**, and the bearing assembly **210** also applies to this aspect, and the same reference numerals will be used for components as in the discussion above. A detailed description of these components is omitted here.

(49) Instead of the bearing **200** being directly thermally connected to the rotor bearing housing **214**, a heat pipe **232** is positioned between the rotor bearing housing **214** and the bearing **200** to transfer heat from the rotor bearing housing **214** to the bearing **200**. In this aspect, the heat pipe **232** has an outer end **234** and an inner end **236**. The outer end **234** is directly attached to the rotor bearing housing **214** and, more specifically, in this aspect, the inner circumferential surface **214c**. The inner end **236** is directly attached to the outer race **206** and, more specifically, in this aspect, the outer circumferential surface **206b**. The heat pipe **232** includes a heat transfer fluid located therein. The heat transfer fluid may be the heat transfer fluids, such as hydrogen, or preferably helium, discussed above with reference to FIG. 5. The heat transfer fluid is heated at the outer end **234** and absorbs heat from the rotor **120** and, more specifically, the rotor bearing housing **214**. Heated heat transfer fluid then flows by natural circulation to the inner end **236** where the heat is transferred to the bearing **200** and, more specifically, the outer race **206**. The heat transfer fluid then continues the natural circulation flow back to the outer end **234**, where the process is repeated. In this aspect, where the heat pipe **232** is between the bearing **200** and the rotor bearing housing **214**, the heat pipe **232** rotates with the rotor **120**. The rotation of the heat pipe **232** thus enhances the heat transfer and flow of the heat transfer fluid between the outer end **234** and the inner end **236**.

(50) In the aspect shown in FIG. 5, the heat pipe **232** includes features to help facilitate heat transfer to and from the heat transfer fluid within the heat pipe **232**. Each of the outer end **234** and the inner end **236** may have a larger cross-sectional area relative to a central section **238** of the heat pipe **232**. Additionally, fins **239** may be formed on each of the inner circumferential surface **214c** and the outer circumferential surface **206b** to further facilitate heat transfer to and from the heat transfer fluid.

(51) FIG. 7 is a cross-sectional view of a bearing assembly **240** that may be used as the high-heat transfer bearing assembly **150**. The bearing assembly **240** is similar to the bearing assembly **220** discussed above with reference to FIG. 5. The same reference numerals will be used for components of the bearing assembly **220** of this aspect that are the same as or similar to the

components of the bearing assembly **220** discussed above. The description of these components above also applies to this aspect, and a detailed description of these components is omitted here. Instead of the rotor bearing housing **224** being directly connected to the rotor **120**, the rotor bearing housing **224** of this aspect is thermally connected to the rotor **120** by a heat pipe, such as the heat pipe **232** discussed above with reference to FIG. **6**. The discussion of the heat pipe **232** above, also applies to this aspect, and the same reference numerals will be used for components as in the discussion above. A detailed description of the heat pipe **232** is omitted here. The cavity **226** may be fluidly connected to the heat pipe **232** to allow the heat transfer fluid to flow therebetween.

(52) Although described with reference to the motor **100**, the bearing **200** and the bearing assemblies **210**, **220**, **230**, and **240** discussed herein may be utilized in other rotating devices, particularly, rotating devices operating at cryogenic temperatures. The bearing **200** and the bearing assemblies **210**, **220**, **230**, and **240** discussed herein can efficiently and effectively transfer heat between from the rotor **120** to the stator **110** of the motor **100** to maintain the windings **112**, **122** at appropriate operating temperatures, particularly at cryogenic temperatures used to maintain the windings at a superconducting temperature.

(53) Further aspects of the present disclosure are provided by the subject matter of the following clauses.

(54) A superconducting motor comprises a rotor including a superconducting winding formed from a superconducting material, a heat sink positioned within the superconducting motor to absorb heat produced by the superconducting motor, and a bearing including an inner race, an outer race, and a rolling element, the bearing rotatably supporting the rotor and thermally connected to each of the rotor and the heat sink to transfer heat from the rotor to the heat sink, the rolling element being formed from a material having, at 25 K, a thermal conductivity of 20 W/mK or more and an ultimate tensile strength of 650 MPa or more.

(55) The superconducting motor of the preceding clause, each of the inner race and the outer race being formed from a material having, at 25 K, a thermal conductivity of 20 W/mK or more and an ultimate tensile strength of 650 MPa or more.

(56) The superconducting motor of any preceding clause, the rolling element being formed from a material having, at 25 K, a thermal conductivity of 100 W/mK or more.

(57) The superconducting motor of any preceding clause, the rolling element being a ball.

(58) The superconducting motor of any preceding clause, one of the outer race and the inner race contacts the rotor.

(59) The superconducting motor of any preceding clause, the rolling element being formed from a copper alloy.

(60) The superconducting motor of any preceding clause, each of the inner race and the outer race being formed from a copper alloy.

(61) The superconducting motor of any preceding clause, further comprising a stator, the bearing rotatably supporting the rotor relative to the stator.

(62) The superconducting motor of any preceding clause, the heat sink being located within the stator.

(63) The superconducting motor of any preceding clause, further comprising a bearing assembly including the bearing, a stator bearing housing connected to the stator, and a rotor bearing housing connected to the rotor, the stator bearing housing opposing the rotor bearing housing with the bearing therebetween.

(64) The superconducting motor of any preceding clause, the stator bearing housing being in direct contact with the heat sink to transfer the heat from the bearing to the heat sink.

(65) The superconducting motor of any preceding clause, the heat sink including a cooling coil through which a coolant flows to absorb heat.

(66) The superconducting motor of any preceding clause, the coolant being liquid hydrogen.

(67) The superconducting motor of any preceding clause, further comprising a bearing assembly

including the bearing and at least one bearing housing, the bearing being located within the at least one bearing housing.

(68) The superconducting motor of any preceding clause, the at least one bearing housing being a rotor bearing housing connected to the rotor, the bearing being thermally connected to the rotor by the rotor bearing housing.

(69) The superconducting motor of any preceding clause, the at least one bearing housing being formed from a copper alloy.

(70) The superconducting motor of any preceding clause, the at least one bearing housing having a cross-sectional area that is a U-shape with two axial end portions extending in a radial direction on either axial side of the bearing, each axial end portion including an inner axial surface abutting an axial surface of one of the inner race and the outer race.

(71) The superconducting motor of any preceding clause, the at least one bearing housing including a cavity formed therein, the cavity being filled with a heat transfer fluid.

(72) The superconducting motor of any preceding clause, the heat transfer fluid consisting essentially of helium.

(73) The superconducting motor of any preceding clause, the at least one bearing housing including a mesh located within the cavity and thermally coupled to a surface of the at least one bearing housing contacting the bearing.

(74) The superconducting motor of any preceding clause, further comprising a bearing assembly including the bearing and a heat pipe having a heat transfer fluid located therein.

(75) The superconducting motor of any preceding clause, the heat transfer fluid consisting essentially of helium.

(76) The superconducting motor of any preceding clause, the heat pipe being positioned between the rotor and the bearing to transfer heat from the rotor to the bearing.

(77) The superconducting motor of any preceding clause, further comprising a rotor bearing housing connected to the rotor, the heat pipe being thermally connected to the rotor by the rotor bearing housing.

(78) The superconducting motor of any preceding clause, the rotor bearing housing being formed from a copper alloy.

(79) Although the foregoing description is directed to the preferred aspects, other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or the scope of the disclosure. Moreover, features described in connection with one aspect may be used in conjunction with other aspects, even if not explicitly stated above.

Claims

1. A superconducting motor comprising: a rotor including a superconducting winding formed from a superconducting material; a stator including a heat sink positioned within the stator to absorb heat produced by the superconducting motor; and a bearing including an inner race, an outer race, and a rolling element, the bearing rotatably supporting the rotor relative to the stator, the bearing thermally connected to each of the rotor and the heat sink to transfer heat from the rotor to the heat sink located within the stator, the rolling element being formed from a material having, at 25 K, a thermal conductivity of 20 W/mK or more and an ultimate tensile strength of 650 MPa or more.

2. The superconducting motor of claim 1, wherein each of the inner race and the outer race is formed from a material having, at 25 K, a thermal conductivity of 20 W/mK or more and an ultimate tensile strength of 650 MPa or more.

3. The superconducting motor of claim 1, wherein the rolling element is formed from a material having, at 25 K, a thermal conductivity of 100 W/mK or more.

4. The superconducting motor of claim 1, wherein one of the outer race or the inner race contacts the rotor.

5. The superconducting motor of claim 1, wherein the heat sink includes a cooling coil through which a coolant flows to absorb heat.
 6. The superconducting motor of claim 1, wherein the rolling element is formed from a copper alloy.
 7. The superconducting motor of claim 1, wherein each of the inner race and the outer race is formed from a copper alloy.
 8. The superconducting motor of claim 1, further comprising a bearing assembly including the bearing, a stator bearing housing connected to the stator, and a rotor bearing housing connected to the rotor, the stator bearing housing opposing the rotor bearing housing with the bearing therebetween.
 9. The superconducting motor of claim 8, wherein the stator bearing housing is in direct contact with the heat sink to transfer the heat from the bearing to the heat sink.
 10. The superconducting motor of claim 1, further comprising a bearing assembly including the bearing and at least one bearing housing, the bearing being located within the at least one bearing housing.
 11. The superconducting motor of claim 10, wherein the at least one bearing housing is a rotor bearing housing connected to the rotor, the bearing being thermally connected to the rotor by the rotor bearing housing.
 12. The superconducting motor of claim 10, wherein the at least one bearing housing is formed from a copper alloy.
 13. The superconducting motor of claim 10, wherein the at least one bearing housing includes a cavity formed therein, the cavity being filled with a heat transfer fluid.
 14. The superconducting motor of claim 13, wherein the heat transfer fluid consists essentially of helium.
 15. The superconducting motor of claim 13, wherein the at least one bearing housing includes a mesh located within the cavity and thermally coupled to a surface of the at least one bearing housing contacting the bearing.
 16. The superconducting motor of claim 10, wherein the at least one bearing housing has a cross-sectional area that is a U-shape with two axial end portions extending in a radial direction on either axial side of the bearing, each axial end portion including an inner axial surface abutting an axial surface of one of the inner race and the outer race.
 17. The superconducting motor of claim 1, further comprising a bearing assembly including the bearing and a heat pipe having a heat transfer fluid located therein.
 18. The superconducting motor of claim 17, wherein the heat transfer fluid consists essentially of helium.
 19. The superconducting motor of claim 17, wherein the heat pipe is positioned between the rotor and the bearing to rotate with the rotor and transfer heat from the rotor to the bearing.
 20. The superconducting motor of claim 17, wherein the heat pipe includes an outer end, an inner end, and a central section, one or both of the outer end and the inner end having a larger cross-sectional area than a cross-sectional area of the central section.
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